A Two-Page Note on Option Pricing

Formula Snip example

1 The Black–Scholes model

Let S_t be the price of a stock that follows a geometric Brownian motion with drift μ and volatility σ. The price of a European call option with strike K and maturity T is

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \tag{1}$$

where r denotes the risk-free interest rate and N is the standard normal distribution function. The two arguments are

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{2}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}. \tag{3}$$

Put–call parity links the call to the put price P through $C - P = S_0 - K e^{-rT}$.

2 Risk and return

The capital asset pricing model states that

$$\mathrm{E}[R_i] = r_f + \beta_i \left(\mathrm{E}[R_m] - r_f \right), \tag{4}$$

where r_f is the risk-free rate and β_i is the sensitivity of asset i to the market. The Sharpe ratio of a portfolio with mean return μ_p and volatility σ_p is

$$SR = \frac{\mu_p - r_f}{\sigma_p}. \tag{5}$$

Asset	β_i	Sharpe ratio
Stock A	0.80	0.42
Stock B	1.20	0.35
Stock C	1.50	0.28

Table 1: Example betas and Sharpe ratios.

3 Discounting

A cash flow X received at time T is worth Xe^{-rT} today. With annual compounding at rate y the present value of a bond paying coupon c for n years and face value F is

$$PV = \sum_{k=1}^{n} \frac{c}{(1+y)^k} + \frac{F}{(1+y)^n}. \tag{6}$$

The continuously compounded return over a period is $\ln(S_T/S_0)$, and its expected value under the risk-neutral measure is $(r - \sigma^2/2)T$.